Lecture 16 (March 10)

Today we will introduce q-analogs. The q can thought of as meaning "quantum", but it is also a variable that these expressions depend on.

Definition (q-analog)**.**

A q-analog of a classical object is a version that depends on some parameter q, such that it approaches the classical object as $q \to 1$.

The most basic example of this is the q-number, which is generally defined as the following (even though there are many potential representations, it appears naturally and has nice properties we will go over):

$$[n]_q = 1 + q + q^2 + \cdots + q^{n-1} = \frac{1-q^n}{1-q}.$$

(Sometimes the subscript $_q$ is removed.)

Next, we have the q-factorial:

$$[n]_q! = [1]_q \cdot [2]_q \cdot [3]_q \cdots [n]_q = \frac{(1-q)(1-q^2)\ldots(1-q^{n-1})(1-q^n)}{(1-q)^n}.$$

For example, $[3]_q! = (1+q)(1+q+q^2) = (1+2q+2q^2+q^3)$. Recall from the problem set that this is exactly the generating function of a Mahonian statistic on S_n:

$$[n]_q! = \sum_{w \in S_n} q^{\mathrm{inv}(w)} = \sum_{w \in S_n} q^{\mathrm{maj}(w)}$$

Naturally, we also have the q-binomial coefficients:

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{[n]_q!}{[k]_q![n-k]_q!}.$$

As an example,

$$\begin{bmatrix}4\\2\end{bmatrix}_q = \frac{[1]_q[2]_q[3]_q[4]_q}{[1]_q[2]_q[1]_q[2]_q} = \frac{(1+q+q^2)(1+q+q^2+q^3)}{1\cdot(1+q)} = 1+q+2q^2+q^3+q^4.$$

Similar to our earlier discussion of Eulerian numbers, we observe some nice properties:

- $\begin{bmatrix}n\\k\end{bmatrix}_q$ is always a polynomial in q, $r_0 + r_1 q + r_2 q^2 + \cdots + r_N q^N$.
- All coefficients r_i are positive integers (called **Gaussian coefficients**).
- The coefficients are symmetric/palindromic; $r_i = r_{N-i} \,\forall\, i$.
- The coefficients are unimodal (they weakly increase up to $r_{\lfloor N/2 \rfloor}$, then decrease).

Also note that the sum of coefficients is $\binom{n}{k}$ (it's just the value at $q = 1$, so it results from the definition of q-analog).

Theorem 1 (Gaussian coefficients)**.**

The q-binomial coefficients $\begin{bmatrix}n\\k\end{bmatrix}_q$ indeed satisfy these properties, with $N = k(n-k)$.

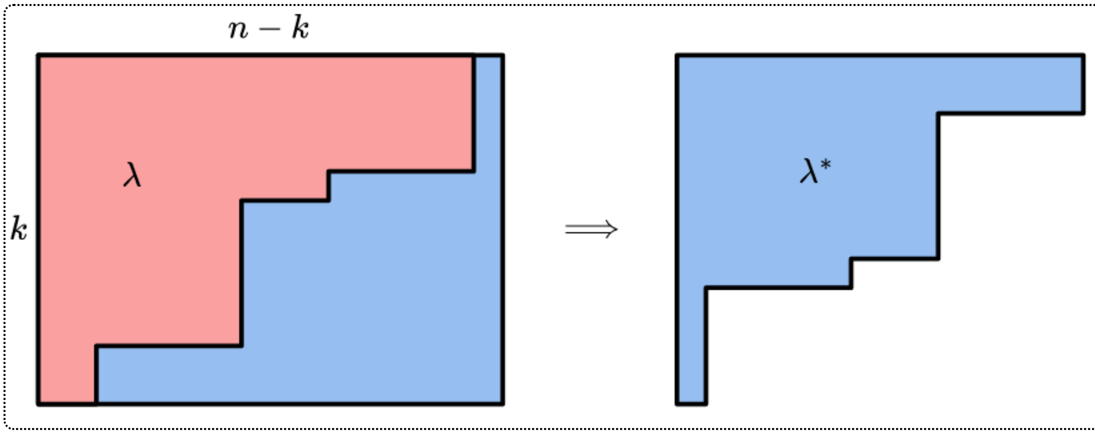
The goal for today's lecture is to prove the first three properties. It would be nice to have a combinatorial interpretation. We will attempt to show the following:

Theorem 2 (Combinatorial interpretation of q-binomial coefficients)**.**

$$\begin{bmatrix}n\\k\end{bmatrix}_q = \sum_{\lambda \subseteq k\times(n-k)} q^{|\lambda|}.$$

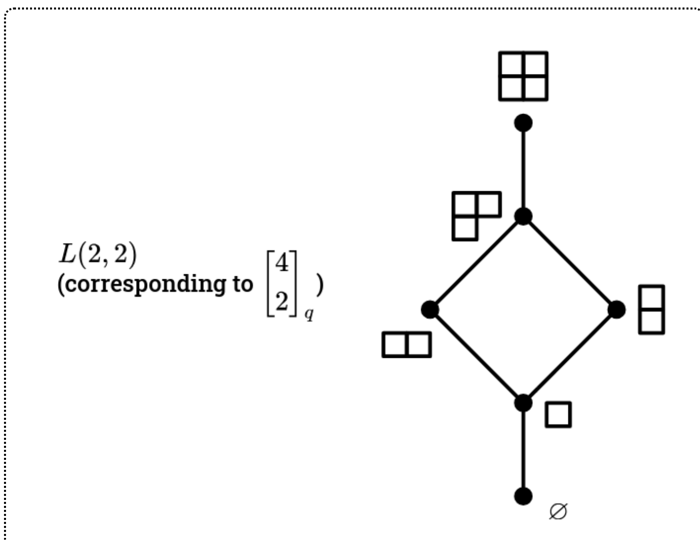
Here, the condition $\lambda \subseteq k \times (n-k)$, means that λ has at most k rows and $n-k$ columns (you can write it as a partition $(n-k \geq \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 0)$.)

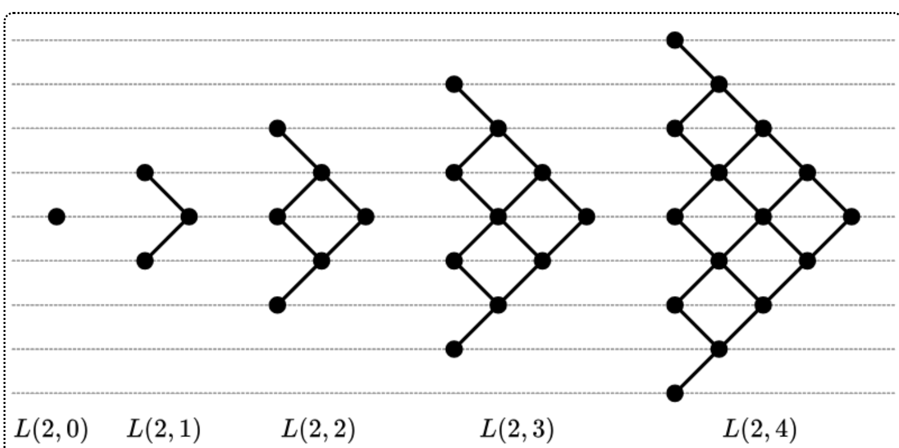
The first three properties easily follow from this theorem, with the palindromic property coming from taking the complement of the Young diagram and rotating (yielding a Young diagram with $k(n-k) - |\lambda|$ squares).



Equivalently, $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is the generating function for the rank numbers of a subset $L(k, n-k)$ of $\mathbb{Y}$ (comprising Young diagrams that fit inside a $k \times (n-k)$ rectangle).

Here are some examples with $k = 2$:





So, we would have $\begin{bmatrix} 6 \\ 2 \end{bmatrix}_q = 1 + q + 2q^2 + 2q^3 + 3q^4 + 3q^4 + 2q^5 + 2q^5 + 2q^6 + q^7 + q^8$.

Note how the Hasse diagrams look like diagonal grids that are cut off.

Proof 1 (by induction on n). We use the following lemma, which can be verified using the definition of $\begin{bmatrix} n \\ k \end{bmatrix}_q$.

> **Lemma 3** (q-Pascal's Identity).
>
> $$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q.$$

This lets us create a "q-Pascal's triangle".

Now define $\begin{bmatrix} n \\ k \end{bmatrix}'_q$ as $\sum_{k \subseteq k \times (n-k)} q^{|\lambda|}$. We can verify base cases $\begin{bmatrix} n \\ 0 \end{bmatrix}'_q = \begin{bmatrix} n \\ n \end{bmatrix}'_q = 1$. Can we prove that this satisfies the same recurrence relation? We will have to split our Young diagrams $\lambda \subseteq k \times (n-k)$ into two subsets, and the way we do this is as follows:

1. The last row is empty, i.e. $\lambda_k = 0$. Then $\lambda \subseteq (k-1) \times (n-k)$, so this case corresponds to the first term.
2. $\lambda_k > 0$, i.e. the first column is full. Then we can remove the first column to obtain $\tilde{\lambda} \subseteq k \times (n-k-1)$. This corresponds to $\begin{bmatrix} n-1 \\ k \end{bmatrix}_{q'}$, and we multiply by q^k to represent the k squares we removed, so this yields the second term.

Therefore, as desired, our combinatorial interpretation does have the same recurrence relation and is therefore the same.

Proof 2 (more conceptual). First assume $q = p^k$ (a power of a prime). If we can prove that our two polynomials are equal for all prime powers, then they must be identically equal.

Then $\mathbb{F}_q$ is the finite field with q elements. (A field is an algebraic structure where you can perform operations like addition/subtraction/multiplication/division; $\mathbb{F}_q$ is unique up to isomorphism.) How can we interpret either side of our equality using $\mathbb{F}_q$?

We will also need $GL_n(\mathbb{F}_q)$, the group of all $n \times n$ matrices with rank n that have elements in $\mathbb{F}_q$.

Lemma 4 (Cardinality of $GL_n(\mathbb{F}_q)$).

The number of elements in $GL_n(\mathbb{F}_q)$ is

$$(q^n - 1)(q^n - q)(q^n - q^2)\dots(q^n - q^{n-1})$$

by selecting one row at a time.

Note how it already looks similar to $[n]_q!$. We will continue this next class.

Lecture 17 (March 12)

We're continuing our Proof 2 from above. First, we have the following definition:

Definition (Grassmannian).

The **Grassmannian** $\mathrm{Gr}(k, n; \mathbb{F})$ for any field $\mathbb{F}$ is the "variety" of all k-dimensional linear subspaces in $\mathbb{F}^n$.

Here, the word "variety" means that it has some structure beyond a set, but for our proof we only need that it is a set. Each element of a Grassmannian is a subspace.

Example (Projective line).

The "projective line" is $\mathbb{P}^1 = \mathrm{Gr}(1, 2; \mathbb{R})$, which comprises every line through the origin. Each one intersects $y = 1$ at a point, except for one where we need to add the point at infinity. So $\mathbb{P}^1$ is a line with P_∞ added.

We can essentially say that $\mathrm{Gr}(k, n; \mathbb{F})$ is the "set" of $k \times n$ matrices of rank k, with elements in $\mathbb{F}$, modulo row operations. A matrix A corresponds to the linear subspace that is its row space (hence why we consider matrices that differ by row operations to be the same).

Recall from 18.06 that a row operation is essentially multiplying by an invertible square matrix on the left ($A \sim A'$ if $A' = BA$ for $B \in GL_k$). So:

$$
\begin{aligned}
\#\,\mathrm{Gr}(k,n;\mathbb{F}_q) &= \frac{\#\{k\times n \text{ matrices of rank } k\}}{\#\{k\times k \text{ matrices of rank } k\}} \\
&= \frac{(q^n-1)(q^n-q)(q^n-q^2)\cdots(q^n-q^{k-1})}{(q^k-1)(q^k-q)(q^k-q^2)\cdots(q^k-q^{k-1})} \\
&= \frac{(q^n-1)(q^{n-1}-1)(q^{n-2}-1)\cdots(q^{n-k+1}-1)}{(q^k-1)(q^{k-1}-1)(q^{k-2}-1)\cdots(q-1)} \\
&= \frac{[n]_q[n-1]_q\cdots[n-k+1]_q}{[k]_q[k-q]_q\cdots[1]_q} \text{ (by dividing every term by } q-1) \\
&= \begin{bmatrix} n \\ k \end{bmatrix}_q .
\end{aligned}
$$

This is pretty important!

Now let's see where we get the other side from. We'll need to recall Gaussian elimination from linear algebra, which lets us use row operations to transform any matrix into a unique one in reduced row echelon form (RREF). In a matrix in RREF,

- the first nonzero entry in every nonempty row is the "pivot" and must be 1
- each column with a pivot cannot have any other nonzero entries.

For example, with $k=4$ and $n=9$, reducing a matrix A via row operations might yield this:

$$
\tilde{A} = \begin{bmatrix}
0 & \boxed{1} & 0 & * & * & 0 & * & 0 & * \\
0 & 0 & \boxed{1} & * & * & 0 & * & 0 & * \\
0 & 0 & 0 & 0 & 0 & \boxed{1} & * & 0 & * \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \boxed{1}
\end{bmatrix}
$$

where the boxed entries are pivots and the $*$s can be any number.

So the size of the Grassmannian is also just the number of matrices in RREF that are of rank k, i.e. have k pivots.

Imagine removing all of the pivot columns. Then it looks like this $k\times(n-k)$ matrix:

$$\tilde{\tilde{A}} = \begin{bmatrix} 0 & * & * & * & * \\ 0 & * & * & * & * \\ 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & * \end{bmatrix}$$

Note that the $*$s are just a reflection of a Young diagram $\lambda \subset k \times (n-k)$, and we can uniquely reinsert the pivot columns to the left of the $*$s! So, the number of matrices in RREF corresponding to this pattern of $*$s is $q^{|\lambda|}$, and summing over all possible patterns we indeed have $\sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|}$, as desired.

Therefore, for all prime powers q, both sides of the equality in Theorem 2 equal the number of elements in $\mathrm{Gr}(k, n; \mathbb{F}_q)$, so the polynomials must be identically equal.

To recap, we have the following two formulas:

$$[n]_q = \sum_{w \in S_n} q^{\mathrm{inv}(w)}$$
$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|}.$$

We want to show that these are actually both cases of the same, more general property. Factorials and binomial coefficients are both cases of multinomial coefficients:

Definition (Multinomial coefficients).

For $k_1 + k_2 + \cdots + k_s = n$,

$$\binom{n}{k_1, k_2, \ldots, k_s} := \frac{n!}{k_1! k_2! \ldots k_s!}$$

This **multinomial coefficient** counts the number of permutations of the multiset comprising k_i copies of i. (A multiset is like a set, but where elements may appear multiple times; this multiset can be denoted as $\{1^{k_1}, 2^{k_2}, \ldots, s^{k_s}\}$).

Note that $s = n$ yields $n!$ and $s = 2$ yields a binomial coefficient.

Naturally, we have q-multinomial coefficients:

$$\begin{bmatrix} n \\ k_1, k_2, \ldots, k_s \end{bmatrix}_q = \frac{[n]_q!}{[k_1]_q! \ldots [k_s]_q!}.$$

Then, we have the following theorem.

Theorem 5 (q-multinomial coefficients).

$$\begin{bmatrix} n \\ k_1, \ldots, k_s \end{bmatrix}_q = \sum_w q^{\mathrm{inv}(w)},$$

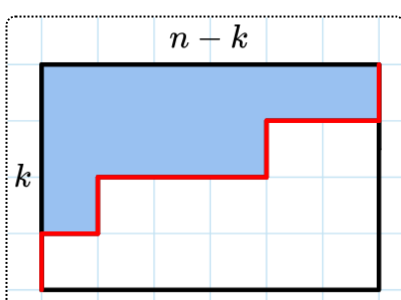
where $w = (w_1, \ldots, w_n)$ is a permutation of the multiset $\{1^{k_1}, 2^{k_2}, \ldots, s^{k_s}\}$, and $\mathrm{inv}(w)$ is the number of pairs $1 \le i < j \le n$ such that $w_i > w_j$.

For example, when computing $\begin{bmatrix} 9 \\ 3,2,4 \end{bmatrix}_q$, $w = (2, 1, 1, 3, 2, 3, 3, 1, 3)$ has 8 inversions.

It's not terribly hard to prove this using induction and a recurrence relation (similar to our Proof 1 from last lecture), or with "partial flag manifolds" (an extension of Grassmannians).

This theorem clearly generalizes the formula we had for $[n]_q!$, but why does the $s = 2$ case yield our formula for $\begin{bmatrix} n \\ k \end{bmatrix}_q$? We need a bijection between Young diagrams $\lambda \subseteq k \times (n-k)$ and permutations of the multiset $\{1^k, 2^{n-k}\}$.

The idea is to look at the path along the boundary from the top-left to the bottom-right. Write a 1 whenever you go up and a 2 whenever you go to the right. For $(n, k) = (10, 4)$ and $\lambda = (6, 4, 1, 0)$, the corresponding permutation is $w = (1, 2, 1, 2, 2, 2, 1, 2, 2, 1)$:



It is not hard to see that each inversion in w corresponds to a square in λ (since each 1 corresponds to a row and 2 to a column, and a square is inside the diagram if and only if the 2 precedes the 1), so $|\lambda| = \mathrm{inv}(w)$ as desired.

Lecture 18 (March 14)

Happy Pi Day! Be sure to observe Pi Moment at 1:59 pm today, right after class ends.

Pi Day tangents

Which is larger, e^π or π^e?

Solution: Take the $\pi \cdot e$-th root of both sides, so you compare $e^{1/e}$ and $\pi^{1/\pi}$. By taking the derivative, $f(x) = x^{1/x}$ has a critical point (local maximum) at $x = e$, so $e^{1/e}$ and therefore e^π is larger. (This is really an e fact, not a π fact).

Mildly more relevant tangent: how many ways are there to permute $-\frac{1}{2}$ objects? The answer is $\left(-\frac{1}{2}\right)!$, which turns out to "be" $\sqrt{\pi}$. The reason is because we have the continuous function $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dz$, we can find $\Gamma(n) = (n-1)!$, and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

In actuality, some of what we cover in class does relate to π:

Limits of q-binomial coefficients

q-binomial coefficients let us take limits on n and k, to yield power series. It would be useless to consider $\lim\limits_{n\to\infty} \binom{n}{k}$, since that's just ∞, but:

$$\lim_{n\to\infty} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(1-q)(1-q^2)\dots(1-q^k)} = \sum_{\substack{\lambda \text{ has at most} \\ k \text{ rows}}} q^{|\lambda|}$$

Also,

$$\lim_{\substack{k\to\infty \\ n-k\to\infty}} \begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{\substack{\lambda \text{ any} \\ \text{Young diag.}}} q^{|\lambda|} = \prod_{n\geq 1} \frac{1}{(1-q^n)}.$$

This lets us find a relation to **partition numbers** $p(n)$, where $p(n)$ is the number of partitions of n. (The above generating function is therefore just $\sum_{n\geq 0} p(n)q^n$.)

n	0	1	2	3	4	5	6	7	8	$\cdots$
$p(n)$	1	1	2	3	5	7	11	15	22	$\cdots$

According to a theorem of Hardy and Ramanujan (1918), as $n \to \infty$,

$$p(n) \sim \frac{1}{4n\sqrt{3}} \cdot e^{\pi\sqrt{2n/3}}$$

(unfortunately this does not yield very accurate approximations of π).

Unimodality of Gaussian coefficients

We can rewrite our sum formula for $\left[{n \atop k}\right]_q$ as

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{i=0}^{N} r_i q^i,$$

where $N = k(n-k)$ and r_i are the rank numbers in Young's lattice (the number of Young diagrams $\lambda \subseteq k \times (n-k)$ with i squares).

Theorem 6 (Sylvester (1978)).

Fix k and n. The rank numbers r_i are unimodal, with

$$r_0 \leq r_1 \leq \cdots \leq r_{\lfloor N/2 \rfloor} \geq \cdots \geq r_N.$$

Proof. Denote by V_i the space of formal linear combinations of Young diagrams $\lambda \subseteq k \times (n-k)$ with $|\lambda| = i$. (This is a subspace of $\mathbb{R}[\mathbb{Y}]$ from previous classes). Note that $\dim(V_{i)}) = r_i$, with a basis comprising all of our individual Young diagrams.

We will use weighted up and down operators (an extension of U and D from last week's classes). U_i will be a linear transformation from V_i to V_{i+1}, such that:

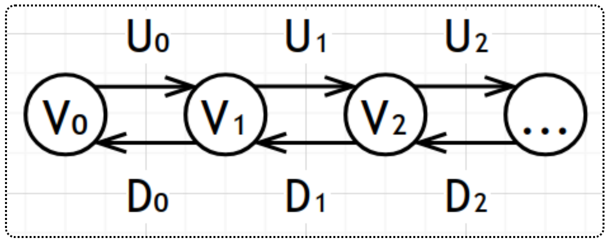
$$U_i\colon \lambda \mapsto \sum_{\mu = \lambda \cup \{a\}} \mathrm{wt}(a) \cdot \mu$$

where a is a single box and wt is a function from squares in $k \times (n-k)$ to $\mathbb{R}_{>0}$.

Similarly, $D_i : V_{i+1} \to V_i$ is

$$\lambda \mapsto \sum_{\mu = \lambda \setminus \{b\}} \mathrm{wt}(b) \cdot \mu.$$

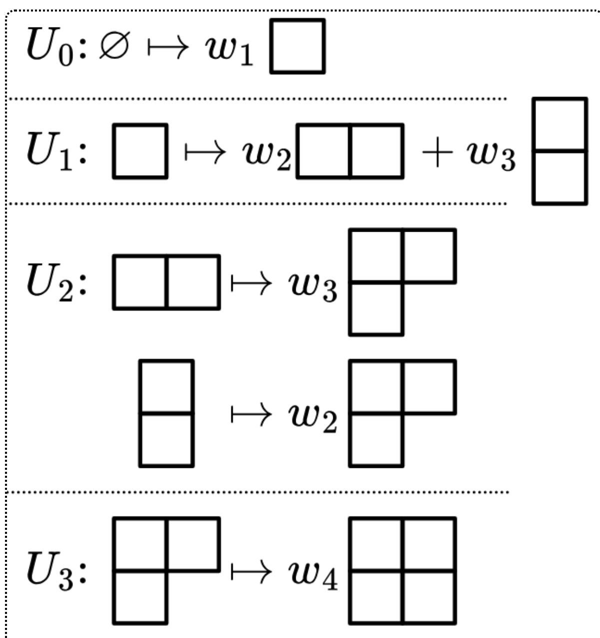
These operators interact with elements of our vector spaces as follows.



They seem scary, but we can essentially just interpret the operators as matrices. For example, with $k = 2, n = 4$, suppose our weights are as follows:

w_1	w_2
w_3	w_4

Then:



and more concretely, we can associate these with the following matrices:

$$U_0 = (w_1) \qquad U_1 = (w_2 \quad w_3) \qquad U_2 = \begin{pmatrix} w_3 \\ w_2 \end{pmatrix} \qquad U_3 = (w_4)$$

In general, U_i is a $r_i \times r_{i+1}$ matrix, and $D_i = U_i^T$ is a $r_{i+1} \times r_i$ matrix.

How do these operators help us? Well, we have a similar crucial lemma to last week!

Lemma 7 (Crucial lemma).

$$D_iU_i = U_{i-1}D_{i-1} + H_i,$$

where H_i is a $r_i \times r_i$ diagonal matrix that sends λ to

$$\left(\sum_{a\in\text{Add}(\lambda)} \text{wt}(a)^2 - \sum_{b\in\text{Remove}(\lambda)} \text{wt}(b)^2\right) \cdot \lambda.$$

$\text{Add}(\lambda)$ and $\text{Remove}(\lambda)$ are the sets of boxes that can be added to or removed from λ, respectively.

The proof of this is the same as last week: you have to add a box a and remove a box b, if $a \neq b$ they cancel out, and otherwise they contribute $\text{wt}(a)^2$ or $\text{wt}(b^2)$.

Now, if we can find a weight function $\text{wt}(a)$ such that

$$\left(\sum_{a\in\text{Add}(\lambda)} \text{wt}(a)^2 - \sum_{b\in\text{Remove}(\lambda)} \text{wt}(b)^2\right) > 0$$

for all λ with i boxes, nice things happen:

- We have that $U_i^TU_i = U_{i-1}U_{i-1}^T + H$.
- For any matrix A, A^TA is symmetric and positive semi-definite (all diagonal entries and determinants of minors are nonnegative)
- H is symmetric and positive definite (from our property of $\text{wt}(a)$)
- So, $U_i^TU_i$ is positive definite.
- That means that its determinant is strictly greater than 0,
- which means that it's invertible and has maximal rank r_i,
- so the rank of U_i is at least r_i.
- Since U_i is a $r_{i+1} \times r_i$ matrix, $r_{i+1} \geq r_i$.

Therefore, we can prove unimodality using such a weight function that works for $|\lambda| < \frac{k(n-k)}{2}$. (The decreasing part immediately follows since the rank numbers are symmetric.)

Construction of the weight function

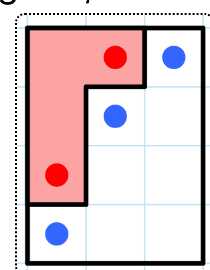
Given a square a in row i and column j, define its "content" $c(a) = j - i$. Then our weight function is

$$\mathrm{wt}(a) = \sqrt{(k + c(a))((n-k) - c(a))} = \sqrt{(k+j-i)(n-k+i-j)}$$

For example, if $k = 4$ and $n = 7$, the values of $\mathrm{wt}(a)^2$ are:

$4 \cdot 3$	$5 \cdot 2$	$6 \cdot 1$
$3 \cdot 4$	$4 \cdot 3$	$5 \cdot 2$
$2 \cdot 5$	$3 \cdot 4$	$4 \cdot 3$
$1 \cdot 6$	$2 \cdot 5$	$3 \cdot 4$

Then, for the following Young diagram,



we get $(6 \cdot 1 + 4 \cdot 3 + 1 \cdot 6) - (5 \cdot 2 + 2 \cdot 5) > 0$.

In fact, we claim the following lemma is true:

Lemma 8 (Explicit value for the above).

For any $\lambda \subseteq k \times (n-k)$,

$$\left(\sum_{a \in \mathrm{Add}(\lambda)} \mathrm{wt}(a)^2 - \sum_{b \in \mathrm{Remove}(\lambda)} \mathrm{wt}(b)^2 \right) = k(n-k) - 2 \cdot |\lambda|.$$

Clearly, this would prove unimodality. Proving this lemma combinatorially will probably appear on the problem set!